



LOST IN ENCODING: COMPARING PROPERTY RETENTION IN TACTILE REPRESENTATION FORMATS

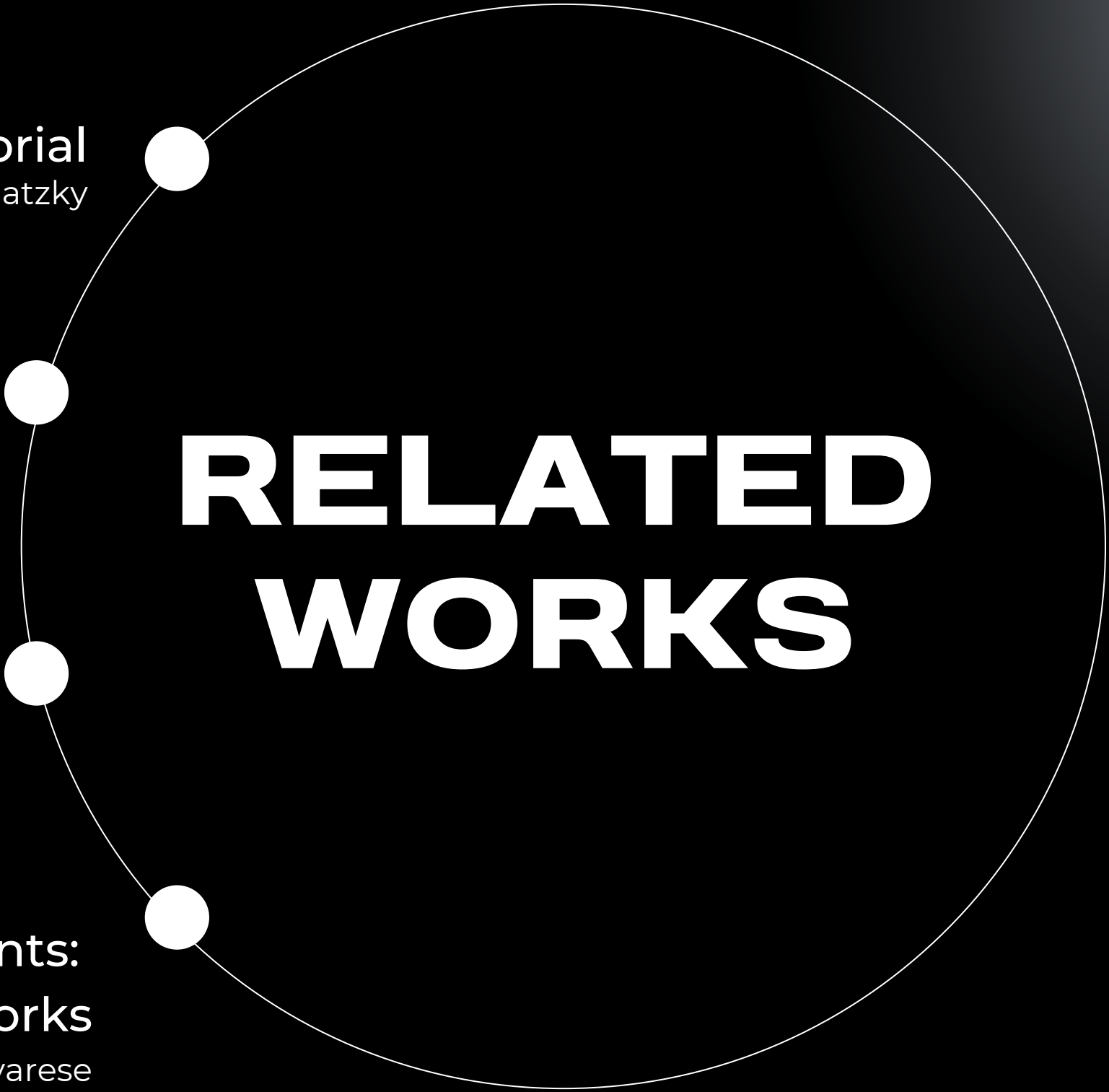
CHARLENE CHEN,
LARA YANG,
HANXUE CHENG,
YUK CHING HE

Haptic Perception: A tutorial
S. J. Lederman and R. L. Klatzky

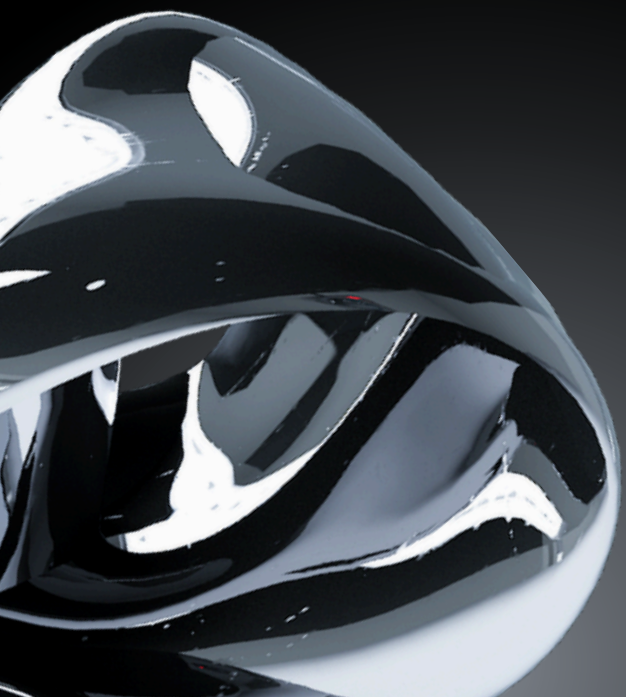
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with Visuotactile Sensing**
Y. Yuan, H. Che, Y. Qin, B. Huang, Z. Yin, K. Lee, Y. Wu, S. Lim, and X. Wang

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C. R. Qi, L. Yi, H. Su, and L. J. Guibas

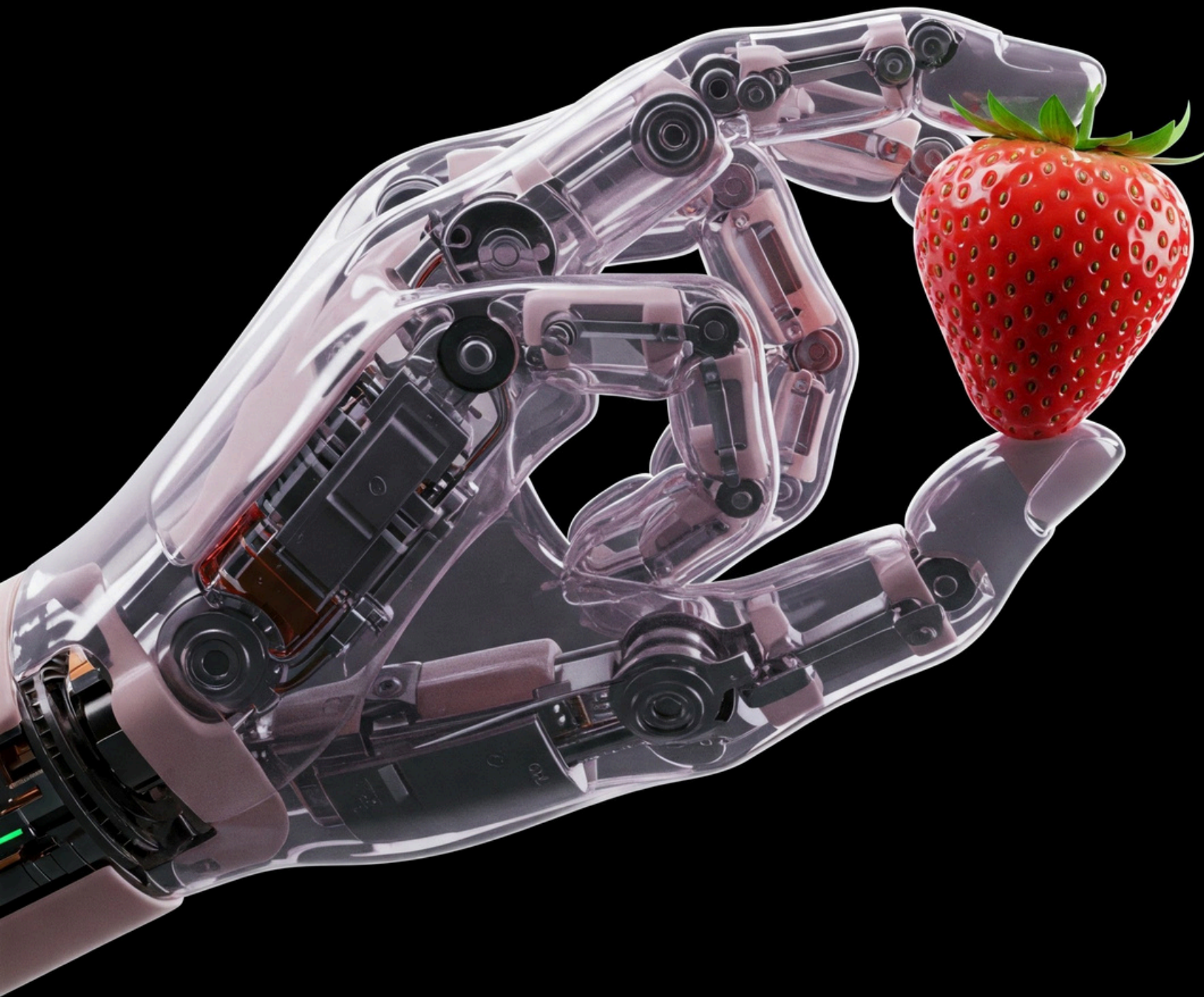
**4D Spatio-Temporal Convents:
Minkowski Convolutional Neural Networks**
C. Choy, J. Gwak, and S. Savarese



RELATED WORKS



CONTEXT



Tactile perception allows robots to understand physical object properties through direct interaction.

Unlike vision-based systems, tactile sensing remains effective under occlusion and captures detailed contact information.

Many tactile classification systems either fuse touch with vision or commit to a single representation type, leaving open how **representation type**, **exploratory procedure**, and **model choice** jointly determine tactile classification performance.

PROBLEM DEFINITION

From **touch alone**, which combination of tactile representation and exploratory procedure best support **feature learning in classification tasks**?

CONTRIBUTIONS



Lara Yanc

- Hardware & ROS Setup
- Dataset Collection & Preprocessing
- Classical ML Models
- Sparse CNN Integration



Charlene Chen

- Hardware & ROS Setup
- Dataset Collection & Preprocessing
- Classical ML Models
- PointNet++ Integration



Hanxue Cheng

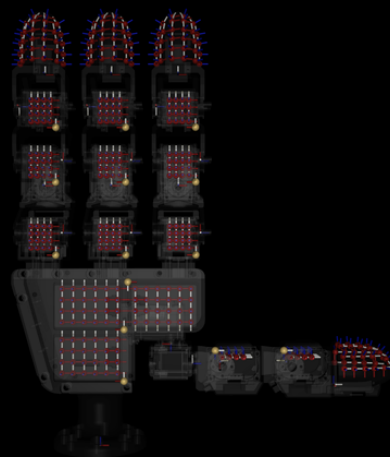
- Point Cloud Generation
- Sensor Calibration & MuJoCo Integration
- External Model Research for PointNet++



Yuk Ching He

- Point Cloud Generation
- Sensor Calibration & MuJoCo Integration
- External Model Research for SparseCNN

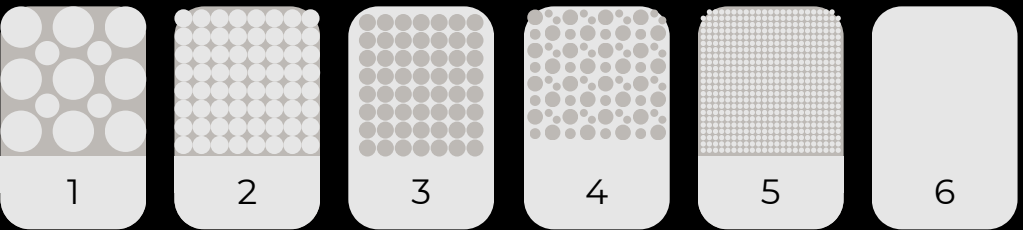
METHODOLOGY



1 Hardware & ROS

- LEAP Robotic Hand
- XELA tactile sensors

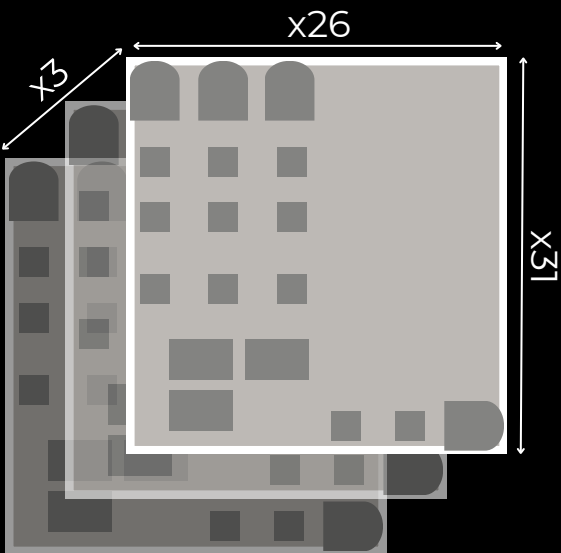
- 6 Exploratory Procedures
- 5 stiffness levels
- 6 texture types
- = 180 recordings, 135,000 frames



2 Data Collection

3 Representation

2D Tactile Image



3D Point Cloud



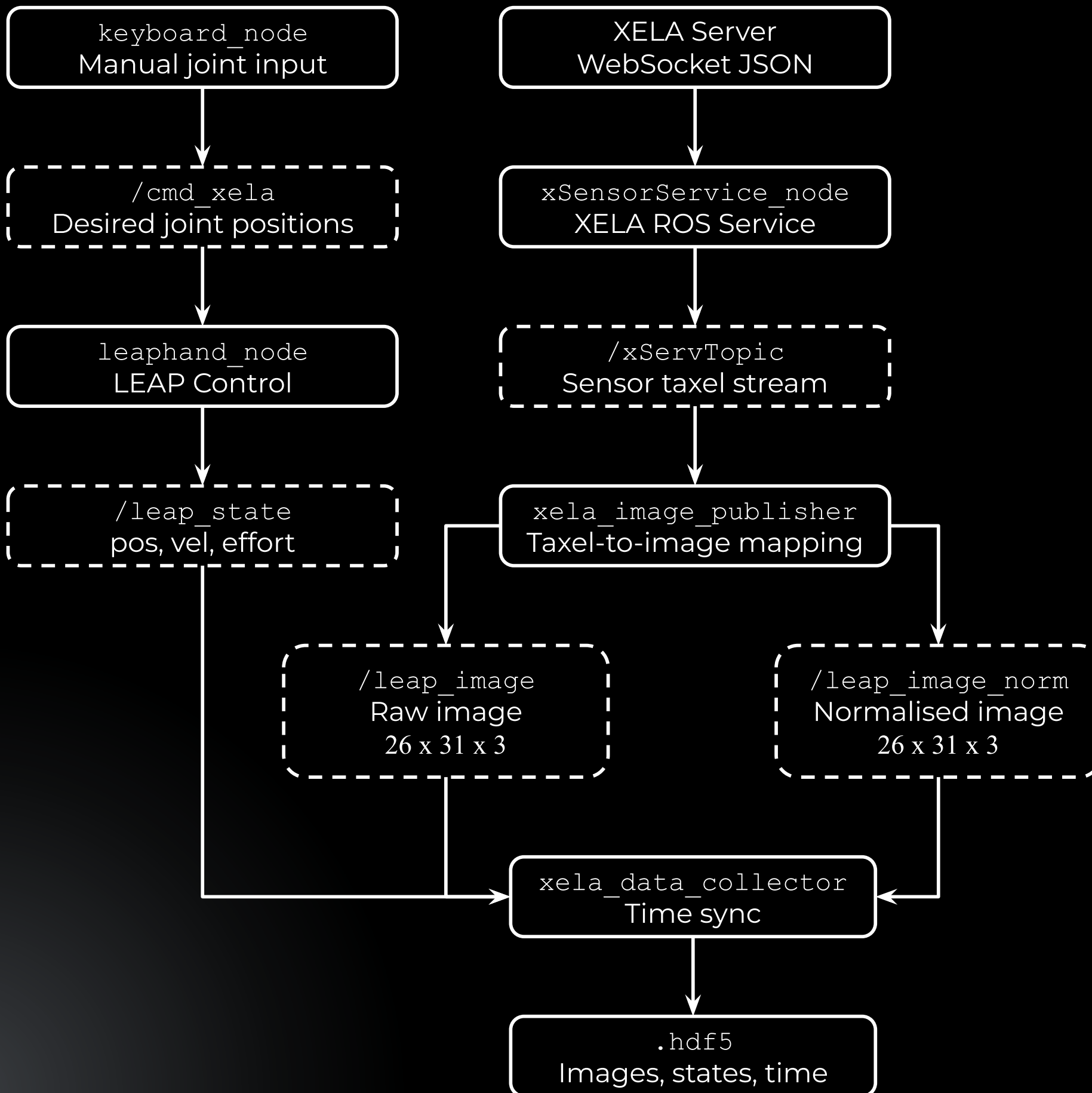
4 Model Implementation

- 4 Machine Learning Models
- 2 Deep Learning Models

5 Evaluation

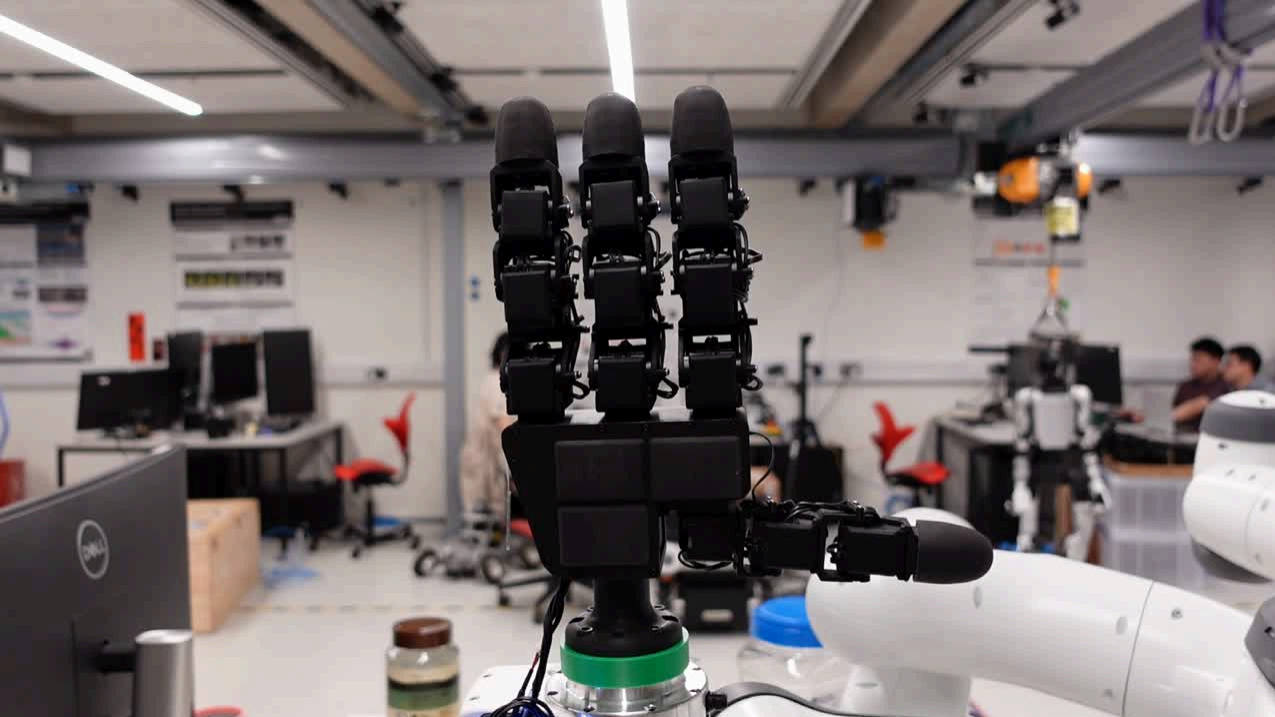
- 3 Variable Experiment
- Stiffness & Texture Classification

ROS2 SYSTEM DESIGN

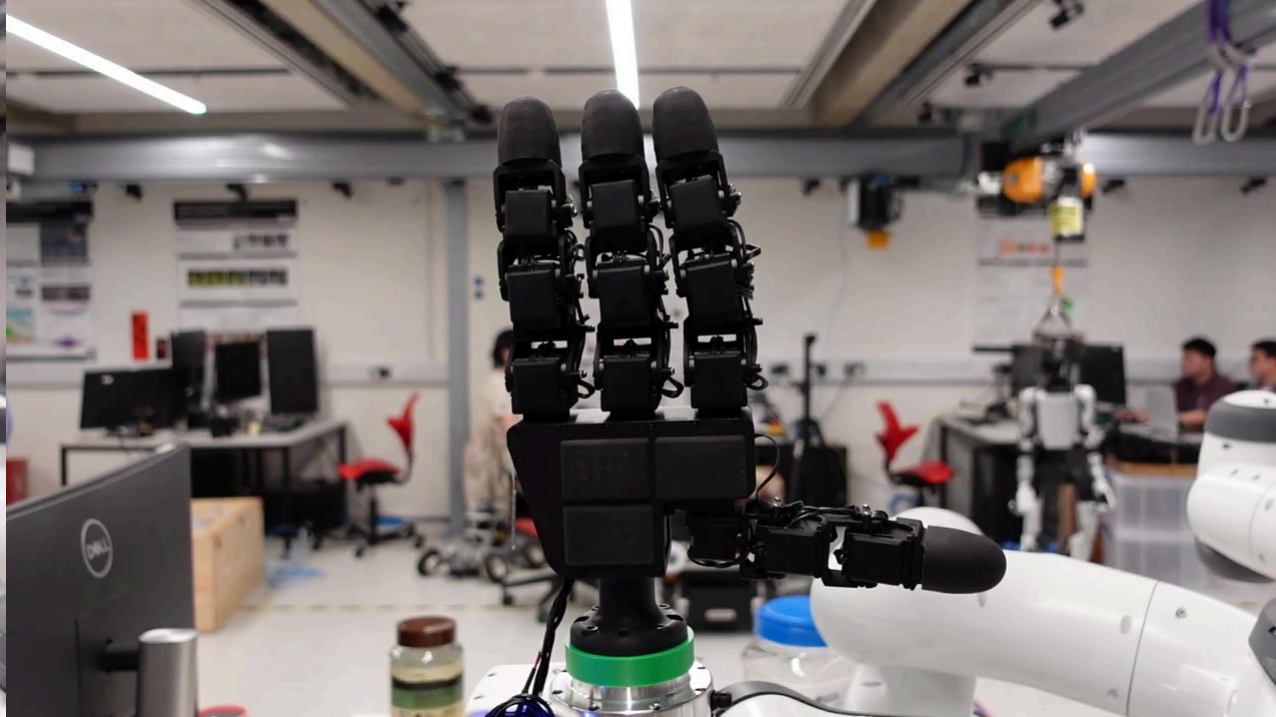


- Execute exploratory procedure (EP)
- Stream tactile + joint data
- Convert taxels into tactile images
- Synchronize and store recordings

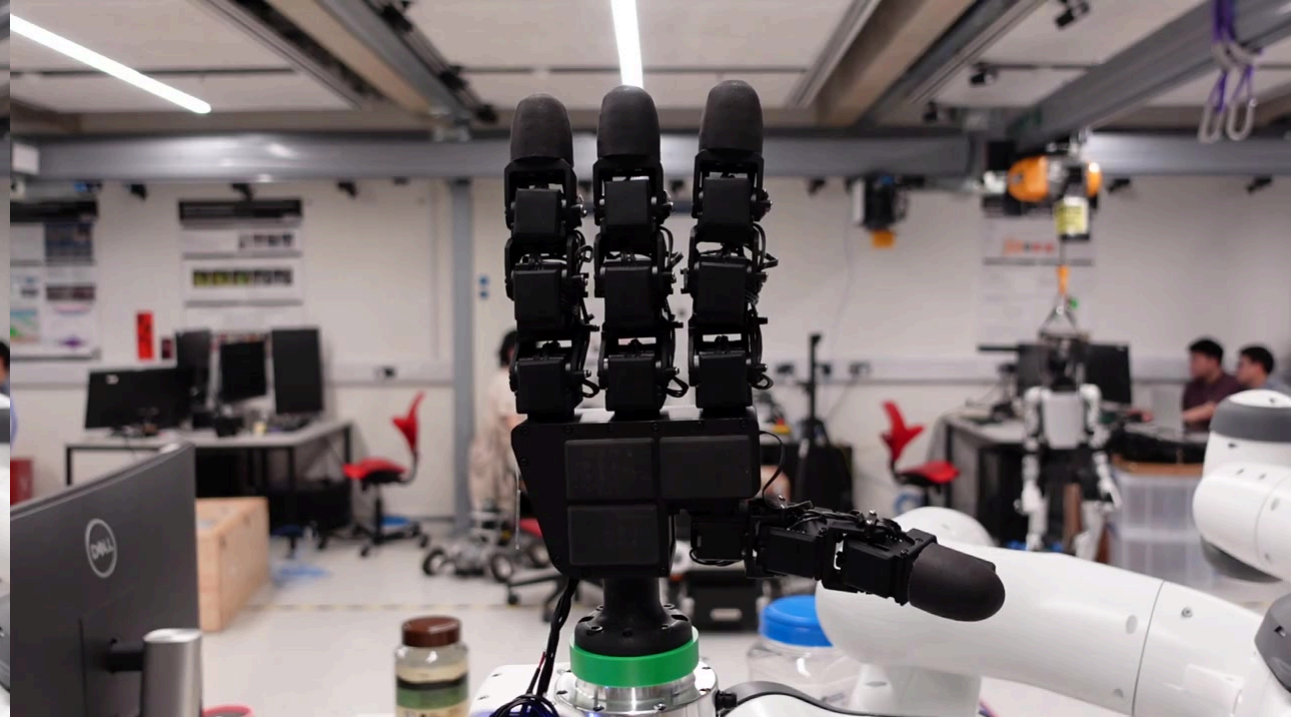
Produces a temporally aligned tactile and proprioceptive dataset for downstream learning.



Flat hand / Stationary
Static



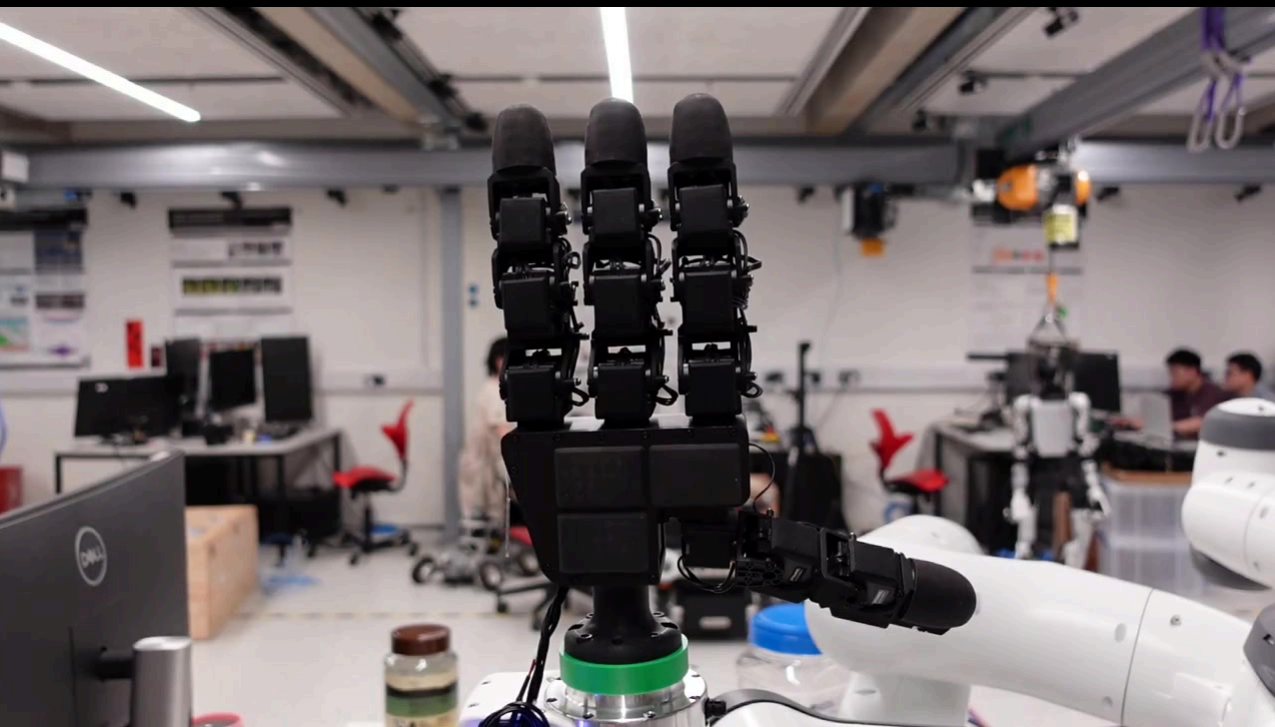
Squeeze
Active



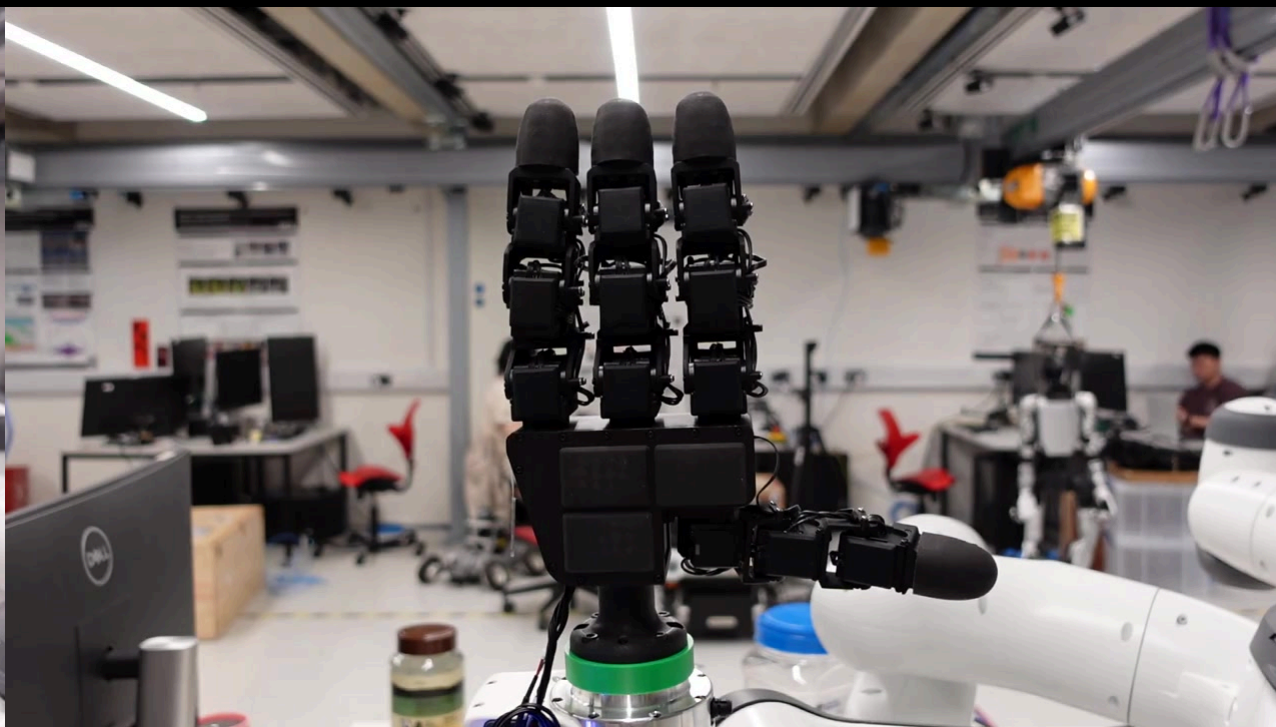
Contour
Active

DATA COLLECTION

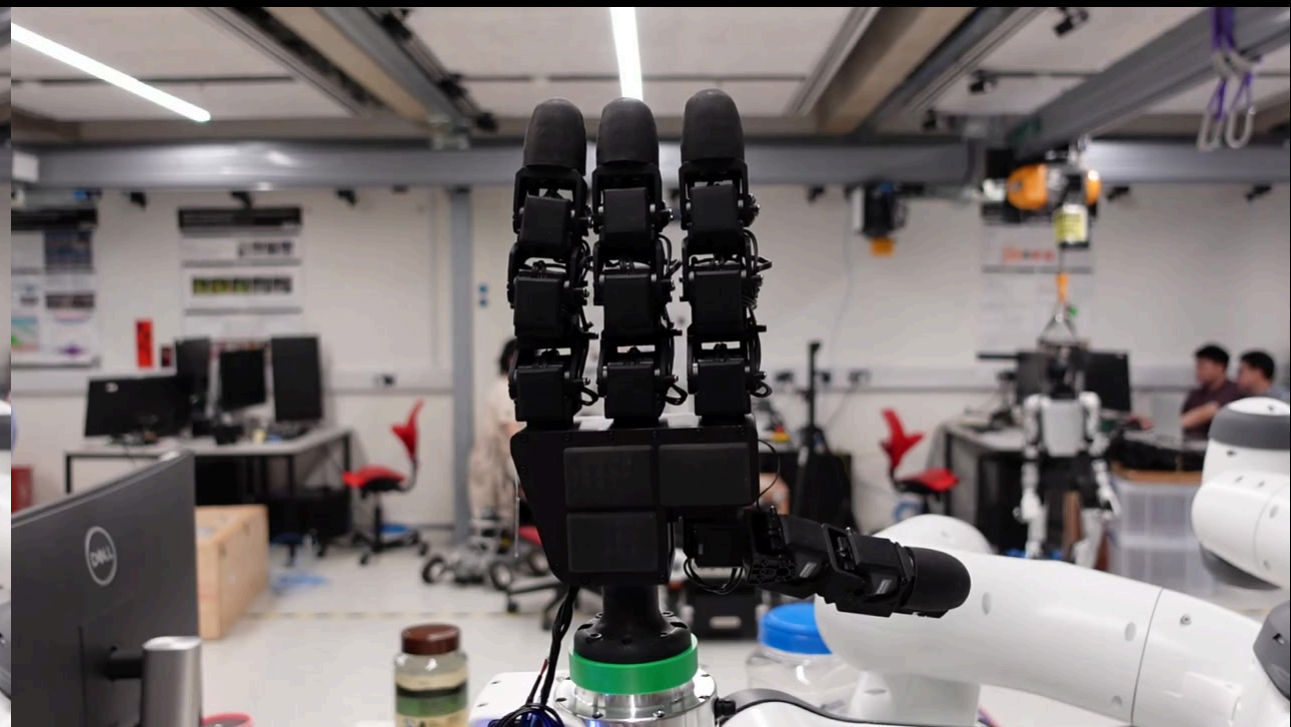
Active
Petting



Active
Enclosure



Active
Enclosure Petting



POINT CLOUD GENERATION



Hardware Modelling in
MuJoCo Simulation

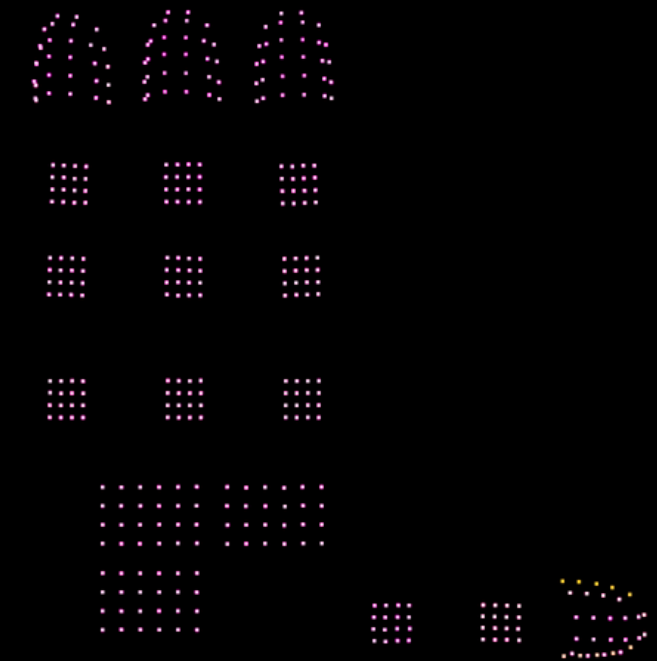
$$\mathcal{F}_{\text{scale}} = \frac{\mathcal{R}_{\text{pointcloud}}}{\mathcal{R}_{\text{taxel}}} = \frac{\mathcal{P}_{\text{max}} - \mathcal{P}_{\text{min}}}{\mathcal{T}_{\text{max}} - \mathcal{T}_{\text{min}}}$$

independent **scale factors** for each
taxel

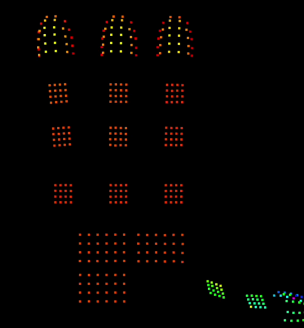
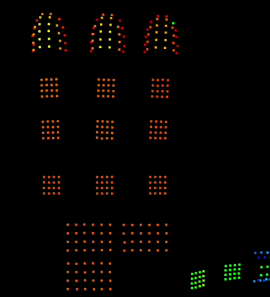
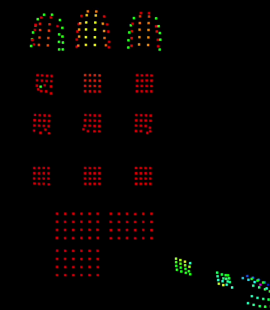
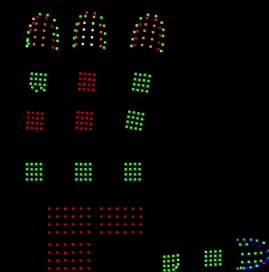
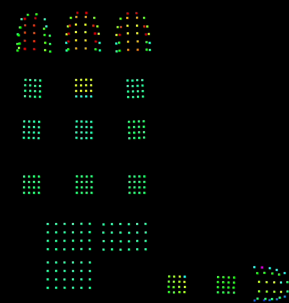
$$\mathcal{D}_{\text{scaled}} = (\mathcal{V}_{\text{sensor}} - \mathcal{V}_{\text{baseline}}) \times \mathcal{F}_{\text{scale}}$$

transform to **local taxel
displacements**

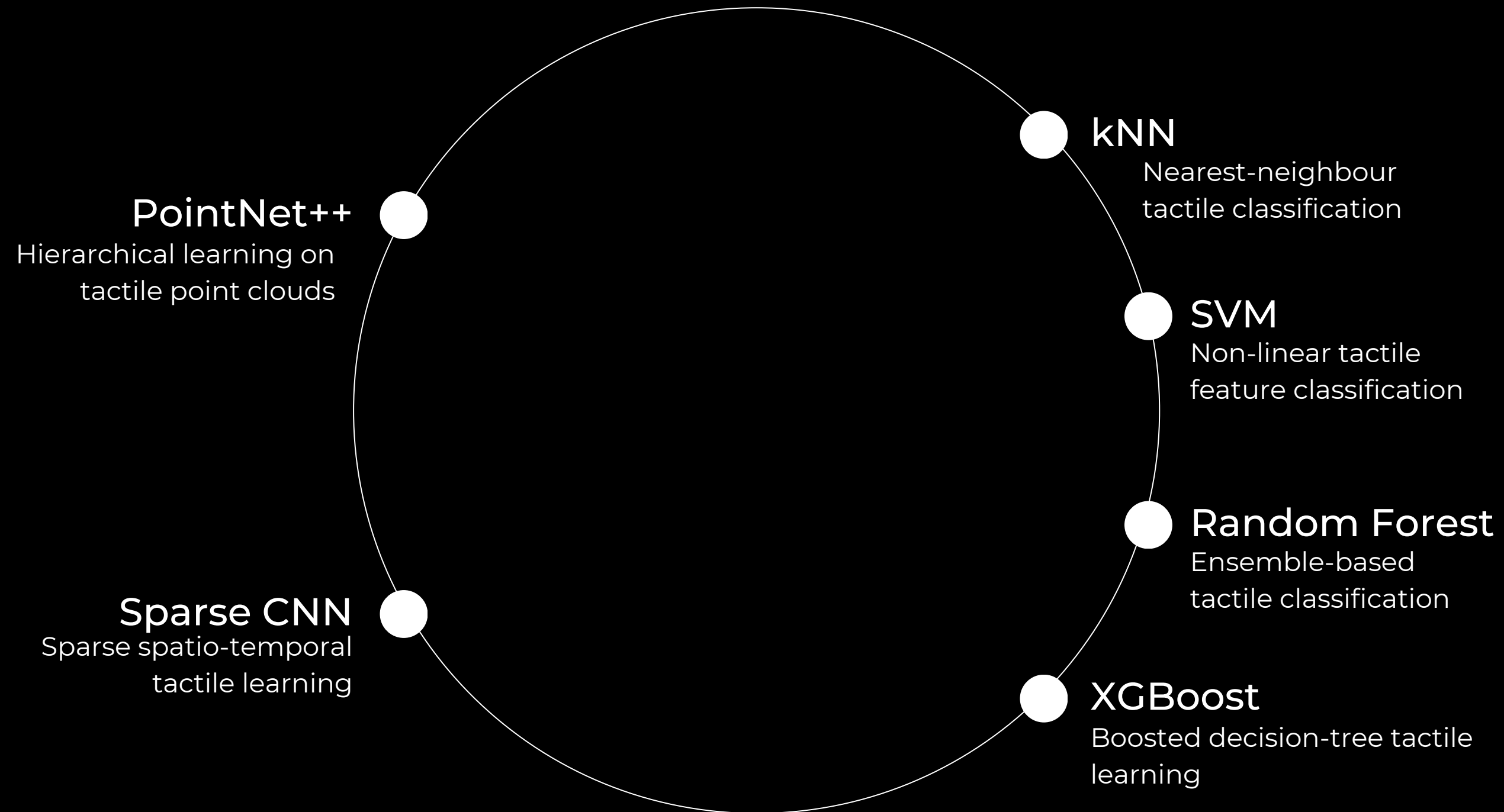
Calibration



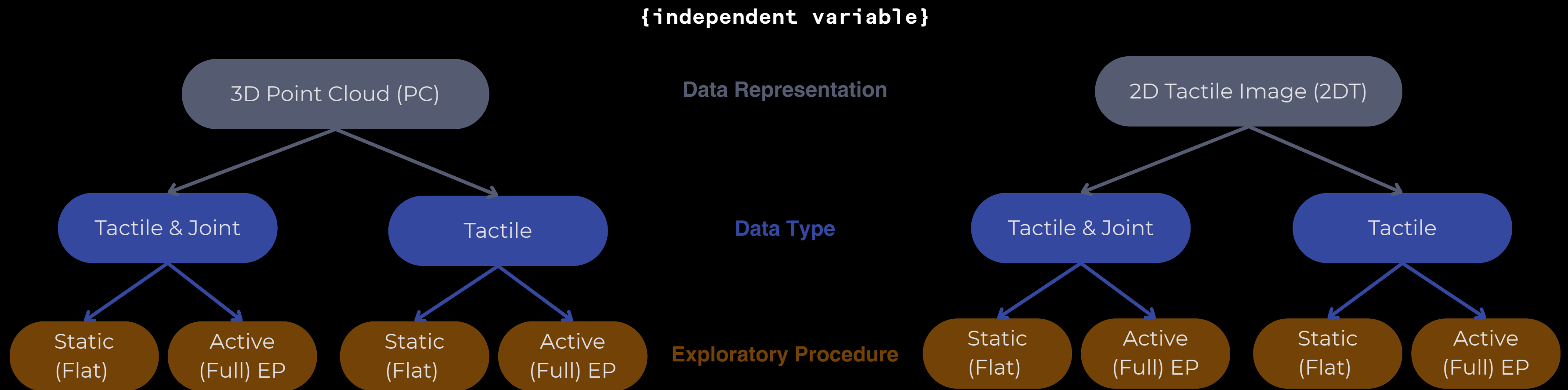
Point Cloud Conversion
through *mj_forward*



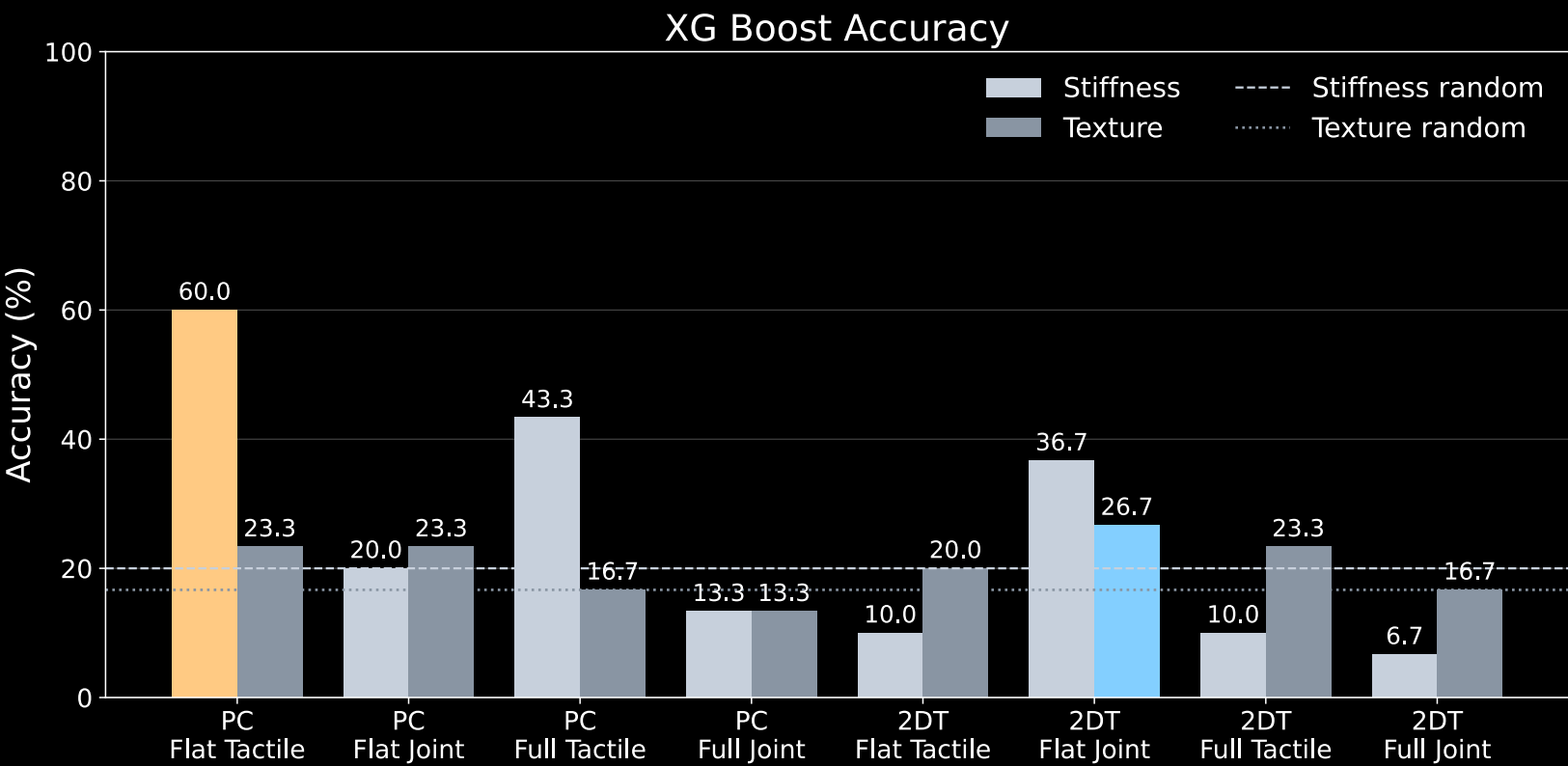
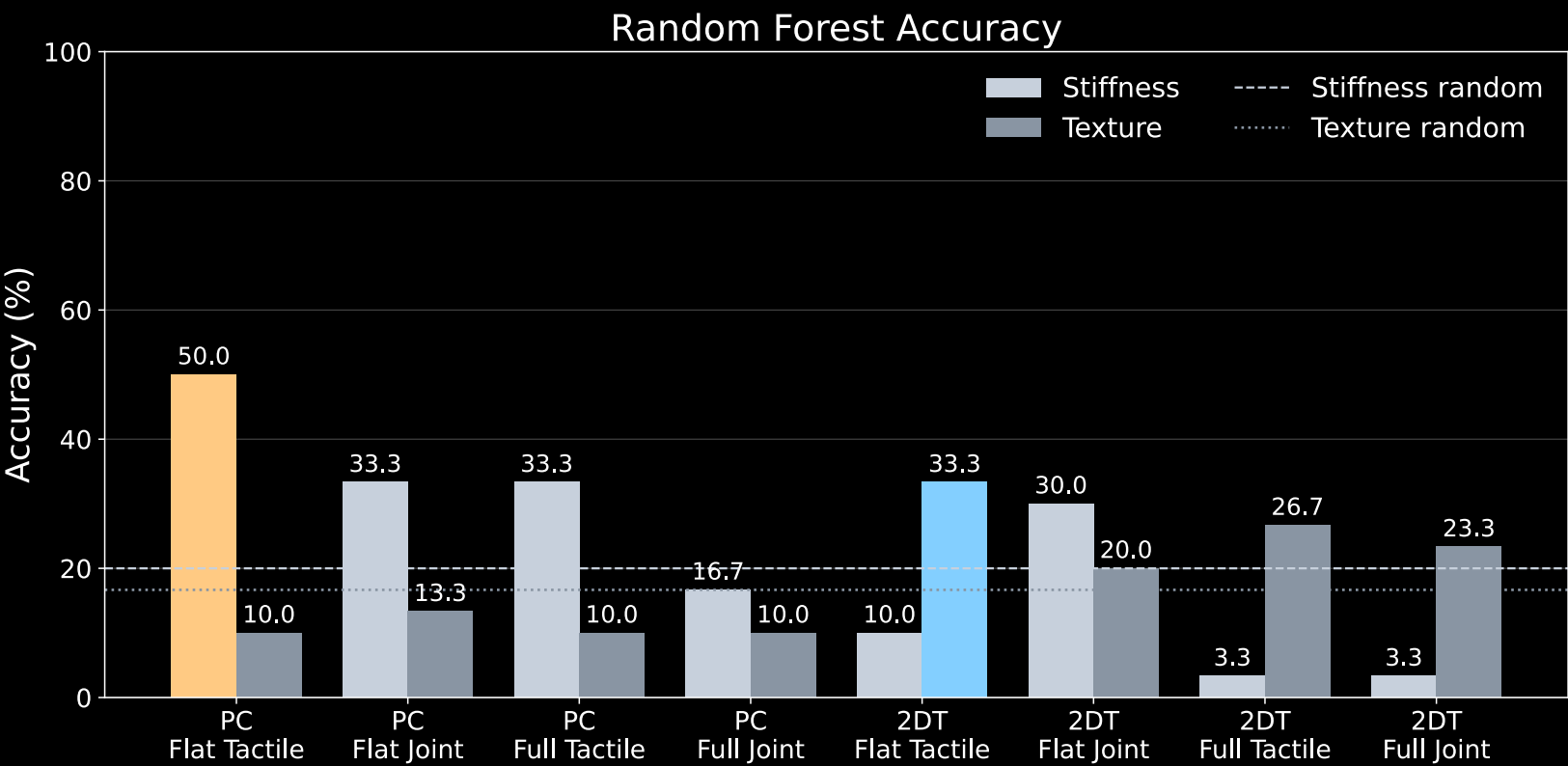
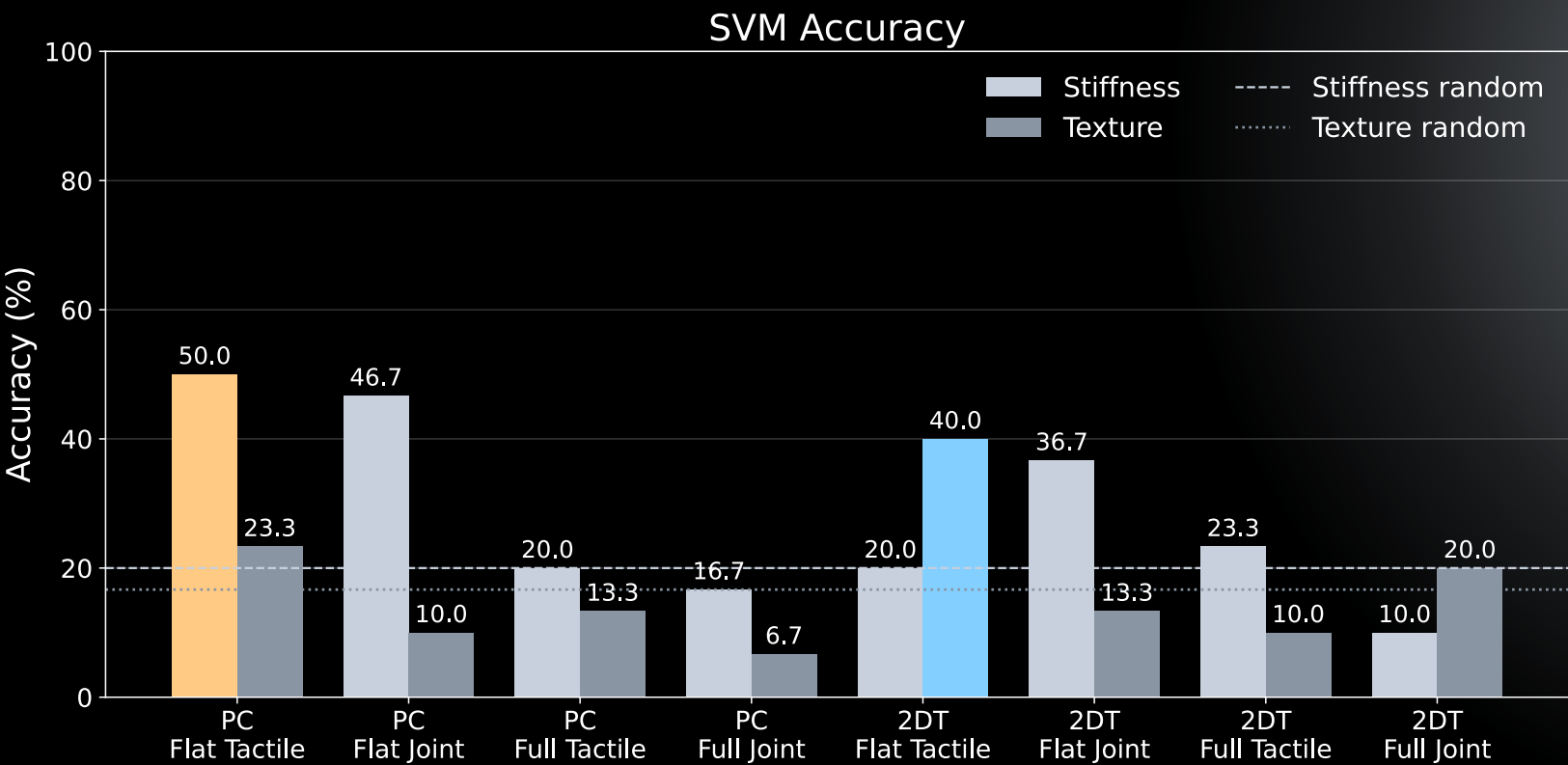
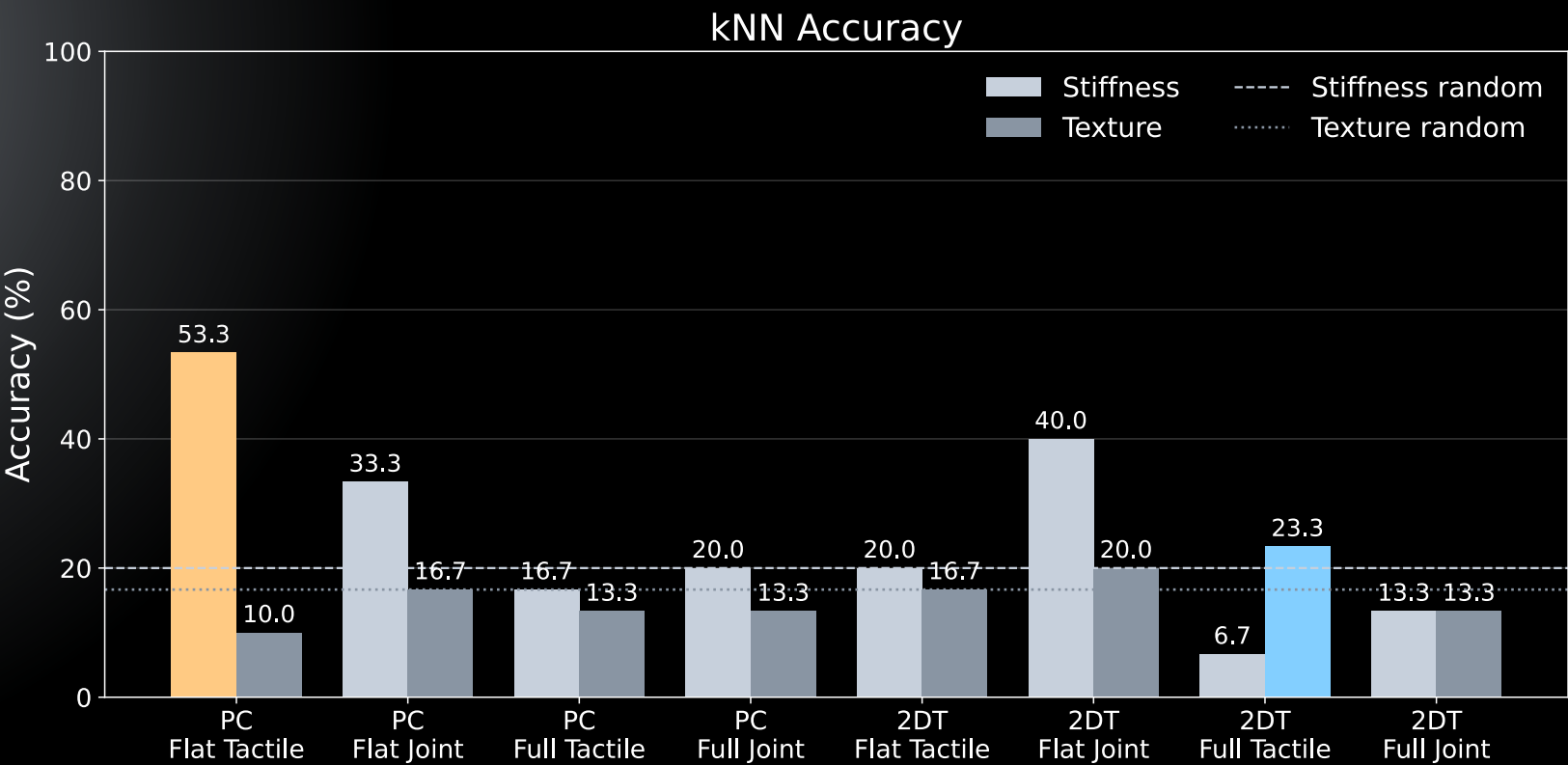
MODEL IMPLEMENTATION



EXPERIMENT SETUP



CLASSICAL ML RESULTS



DEEP LEARNING RESULTS

Model Type	Representation	EP + Data Type	Classification Category	Acc. (%)	Gain* (%)
kNN	Point Cloud	Flat Tactile	Stiffness	53.33	33.33
kNN	2D Tactile Image	Flat Tactile	Texture	23.33	6.67
Random Forest	Point Cloud	Flat Tactile	Stiffness	50.00	30.00
Random Forest	2D Tactile Image	Flat Tactile	Texture	33.33	16.67
SVM	Point Cloud	Flat Tactile	Stiffness	50.00	30.00
<i>SVM</i>	<i>2D Tactile Image</i>	<i>Flat Tactile</i>	<i>Texture</i>	<i>40.00</i>	<i>23.33</i>
<i>XG Boost</i>	<i>Point Cloud</i>	<i>Flat Tactile</i>	<i>Stiffness</i>	<i>60.00</i>	<i>40.00</i>
XG Boost	2D Tactile Image	Flat Tactile + Joint	Texture	26.67	10.00
PointNet++	Point Cloud	Flat Tactile + Joint	Both	(T) 19.78 (S) 17.56	-0.22 0.89
Sparse CNN	2D Tactile Image	Flat Tactile + Joint	Both	(T) 27.78 (S) 38.89	7.78 22.22

*Gain is defined as the difference between classification accuracy and random baseline accuracy.



The project involved **hardware integration and calibration** challenges, limited **dataset size**, and additional complexity in generating tactile-proprioceptive **point cloud representations** through the MuJoCo simulation pipeline.

CHALLENGES & TAKEAWAYS

The project showed the importance of **parallel development**, careful **model selection**, and a well-structured **experimental setup**.

CONCLUSION & FUTURE WORK

Tactile perception enables robots to understand physical object properties through direct interaction rather than vision alone.

Our results showed that tactile **representation choice strongly affects classification performance**, with point clouds performing better for stiffness estimation and tactile images performing better for texture recognition.



THANK YOU

FULL PAPER:

**Lost in Encoding: Comparing Property Retention in
Tactile Representation Formats**

GITHUB REPOSITORY:

<https://github.com/charlene-chenn/tactile-intelligence>